**Highlighted component indicates the custom wrist-roll joint addition.*

# OMX-6DOF-R1

6-DOF Upgraded Edition

**Full Orientation. Designed for research and educational applications.**

The upgraded OMX-6DOF-R1 features a custom 6-DOF configuration with an integrated wrist-roll joint, enabling full spatial orientation. With a compact footprint and an open-source control stack, it provides a cost-effective solution for robotic manipulation, rapid prototyping, and educational applications.

## Key Features

- Full 6-DOF spatial orientation capability
- Native out-of-the-box ROS 2 integration
- Lightweight, modular, and upgradeable design
- Built-on industrial Dynamixel smart servos
- Cost-effective platform for advanced R&D

## Modifications vs. Original

- **Hardware:** Added 1 DOF (wrist-roll joint)
- **Actuation:** J5 utilizes XL430-W250-T servo
- **Software:** Updated URDF & MoveIt kinematics

## Applications

- Pick-and-place automation testing
- Algorithm benchmarking & motion planning
- University robotics education and training

## Performance

| | |
|---|---|
| Max TCP speed | 0.30 m/s |
| Max joint speed | 275°/s |
| Mounting Options | Tabletop or fixed base |
| Noise Level | ¡ 60 dB(A) |
| Orientational Accuracy | ± 0.647 ° |
| Positional Accuracy | ± 0.427 mm |

## General Specification

| | |
|---|---|
| Payload | 0.3 kg (at max reach) |
| Reach | 342 mm |
| Degrees of Freedom | 6 |
| Configuration | Articulated Manipulator |
| Repeatability | ±2.051 mm |
| Footprint | Ø150 mm |

*Payload includes end-effector weight. Reach defined as maximum distance from base to TCP.

## Joint Specifications

| Axis | Working Range | Max Speed | Peak Torque |
|---|---|---|---|
| Axis1 | $-180°$ to $180°$ | 275°/s | 4.1 Nm |
| Axis2 | $-86°$ to $86°$ | 275°/s | 4.1 Nm |
| Axis3 | $-86°$ to $80°$ | 275°/s | 4.1 Nm |
| Axis4 | $-97°$ to $113°$ | 275°/s | 4.1 Nm |
| Gripper Left Axis | $-1°$ to $1°$ | 275°/s | 4.1 Nm |
| Gripper Right Axis | $-1°$ to $1°$ | 275°/s | 4.1 Nm |

*Gripper provides additional actuation DOF. Torque values are derived from manufacturer datasheets.

## Control System

| | |
|---|---|
| Supported OS | Ubuntu 22.04 / 24.04 |
| Middleware | ROS 2 (Humble / Jazzy) |
| Motion Planning | MoveIt 2 (OMPL) |
| Controller | ros2_control |
| Simulation | Gazebo (Ignition) |
| Programming | Python, C++ APIs |
| Interface | USB / Serial (via controller) |

## Actuators

| | |
|---|---|
| Actuator Type | DYNAMIXEL Smart Servo |
| Motor Models | XM430-W350-T / XL430-W250-T |
| Control Mode | Position / Velocity / Current |
| Communication | TTL Multi-Drop Bus |
| Feedback | Absolute Contactless Encoder |

## Electrical Specifications

| | |
|---|---|
| Input voltage | 12 V DC |
| Power supply | 12 V, 5 A adapter |
| Typical power | 35 W |
| Maximum power | 120 W (peak load) |
| Digital I/O | Available via external controller |

## Physical & Kinematics

| | |
|---|---|
| Materials | grey, red |
| Cable length | 1 m |
| Total Weight | $\approx$ 0.59 kg |
| IP Rating | IP20 |
| Operating Temp | 0°C to 45°C |
| Humidity | 5% to 85% (non-condensing) |

## Safety & Compliance

| | |
|---|---|
| Safety rating | Uncertified (Research use only) |
| Architecture | Software-limited force/torque feedback (PL-b equivalent) |
| Protection | Configurable joint, velocity, and torque limits |
| E-Stop | Software-triggered (external hardware optional) |

## Known Limitations

- Not suitable for industrial payloads or high-cycle loads
- Limited stiffness due to 3D-printed PLA structure
- Accuracy subject to standard Dynamixel gear backlash

## Documentation & Resources

**GitHub Repository**
Access source code, ROS 2 packages, setup guides, and updates.
github.com/KMTI-ROBOPARADIGM/omx-wrist-roll-original

## Package Contents

- 6-DOF Robotic Arm & Gripper
- 12V Power Adapter
- Communication Cable
- Mounting Hardware
- Quick start guide / documentation

## Contact & Origin



| | |
|---|---|
| Lab/Entity | **AutoCert Lab** |
| Contact | research@domain.edu |
| Location | India |